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Part A: Multiple choice (Select the one that is best in each case. 1 point/question)

1. Which of the following solutions will be the best buffer at a pH of 9.26? (K_a for CH_3COOH is 1.8×10^{-5}; K_b for NH_3 is 1.8×10^{-5}.)

A) 0.20 *M* NH_3 and 0.20 *M* NH_4Cl

B) 5.0 *M* CH_3COOH and 5.0 *M* NH_4Cl
C) 0.20 *M* CH_3COOH and 0.20 *M* CH_3COONa
D) 5.0 *M* NH_3 and 5.0 *M* NH_4Cl
E) 5.0 *M* CH_3COOH and 5.0 *M* NH_3

2. If 30 mL of 5.0×10^{-4} *M* $Ca(NO_3)_2$ is added to 70 mL of 2.0×10^{-4} *M* NaF, will a precipitate form? (K_{sp} of $CaF_2 = 4.0 \times 10^{-11}$)

A) No, because Q is less than K_{sp}.

B) No, because Q is greater than K_{sp}.

C) Not enough information is given.

D) Yes, because Q is less than K_{sp}.
E) Yes, because Q is greater than K_{sp}.

3. Consider the titration of 100.0 mL of 0.250 *M* aniline ($K_b = 3.8 \times 10^{-10}$) with 0.500 *M* HCl. Calculate the pH of the solution at the equivalence point.

A) 8.70 B) 2.68 C) 11.62 D) – 0.85 E) none of these

4. The two salts AgX and AgY have very similar solubilities in water. The salt AgX is much more soluble in acid than is AgY. What can be said about the relative strengths of the acids HX and HY?

A) HY is stronger than HX.

B) Nothing can be said about their relative strengths.
C) HX is stronger than HY.
D) The acids have equal strengths.

E) None of these statements is correct.

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5. Given the following values of equilibrium constants:

$Cu(OH)_2(s) \rightleftharpoons Cu^{2+}(aq) + 2OH^-(aq)$ $K_{sp} = 1.6 \times 10^{-19}$

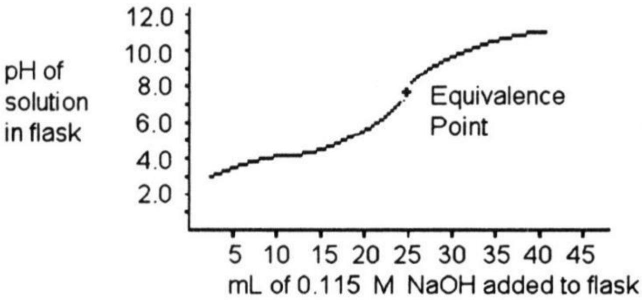
$Cu(NH_3)_4^{2+}(aq) \rightleftharpoons Cu^{2+}(aq) + 4NH_3(aq)$ $K = 1.0 \times 10^{-13}$

What is the value of the equilibrium constant for the following reaction?

$$Cu(OH)_2(s) + 4NH_3(aq) \rightleftharpoons Cu(NH_3)_4^{2+}(aq) + 2OH^-(aq)$$

A) 1.6×10^{-19} B) 1.6×10^{-6} C) 6.2×10^{31} D) 1.0×10^{13} E) 1.6×10^{-32}

6. A 25.0 mL sample of a solution of an unknown compound is titrated with a 0.115 *M* NaOH solution. The titration curve below was obtained. The unknown compound is ________.



A) a weak acid

B) a strong base

C) a strong acid

D) a weak base

E) neither an acid nor a base

7. Indicate whether each statement is true?

A) A reaction that is spontaneous in one direction will be nonspontaneous in the reverse direction under the same reaction conditions.

B) All spontaneous processes are fast.

C) All spontaneous processes are exothermic.

D) An isothermal process is one in which the system loses no heat.

E) The maximum amount of work can be accomplished by an irreversible process rather than a reversible one.

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8. A reversible process is one that ________.
   A) must be carried out at high temperature

   B) happens spontaneously

   C) is spontaneous in both directions

   D) must be carried out at low temperature

   E) can be reversed with no net change in either system or surroundings

9. Which one of the following processes produces a decrease of the entropy of the system?
   A) dissolving potassium chloride in water

   B) sublimation of iodine

   C) dissolving oxygen in water

   D) boiling of ethanol

   E) explosion of nitroglycerine

10. When a system is at equilibrium, ________.
    A) the reverse process is spontaneous but the forward process is not

    B) the forward and the reverse processes are both spontaneous

    C) the forward process is spontaneous but the reverse process is not

    D) the process is not spontaneous in either direction

    E) both forward and reverse processes have stopped

11. The entropy change accompanying any process is given by the equation:
    A) $\Delta S = k \ln W_{final}$

    B) $\Delta S = k\, W_{final} - k\, W_{initial}$

    C) $\Delta S = k \ln(W_{final} / W_{initial})$

    D) $\Delta S = k_{final} - k_{initial}$

    E) $\Delta S = W_{final} - W_{initial}$

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12. For the reaction

$$C_2H_6\ (g) \rightarrow C_2H_4\ (g) + H_2\ (g)$$

ΔH^o is +137 kJ/mol and ΔS^o is +120 J/K · mol. This reaction is ________.

A) spontaneous only at low temperature

B) nonspontaneous at all temperatures

C) spontaneous only at high temperature

D) spontaneous at all temperatures

13. Balance the following half-reaction occurring in acidic solution.
$NO_3^-(aq) \rightarrow NH_4^+(aq)$

A) $NO_3^-(aq) + 10\ H^+(aq) + 8\ e^- \rightarrow NH_4^+(aq) + 3\ H_2O(l)$

B) $NO_3^-(aq) + 7\ H_2O(l) + 8\ e^- \rightarrow NH_4^+(aq) + 10\ OH^-(aq)$
C) $NO_3^-(aq) + 10\ H^+(aq) \rightarrow NH_4^+(aq) + 3\ H_2O(l) + 10\ e^-$
D) $NO_3^-(aq) + 8\ e^- \rightarrow NH_4^+(aq) + 8\ H^+(aq) + 3\ H_2O(l)$
E) $NO_3^-(aq) + 10\ H^+(aq) \rightarrow NH_4^+(aq) + 3\ H_2O(l)$

14. The following reactions take place in a lead-acid battery.
Discharging: $Pb(s) + PbO_2(s) + 4\ H^+(aq) + 2\ SO_4^{2-}(aq) \rightarrow 2\ PbSO_4(s) + 2\ H_2O(l)$
Recharging: $2\ PbSO_4(s) + 2\ H_2O(l) \rightarrow Pb(s) + PbO_2(s) + 4\ H^+(aq) + 2\ SO_4^{2-}(aq)$
Which of the following statements is true?
A) The concentration of H_2SO_4 increases as the battery discharges.
B) Pb is formed at the anode during discharge.
C) Pb is formed at the cathode during recharging.
D) The mass of Pb decreases during recharging.
E) The mass of $PbSO_4$ remains constant during recharging and discharging.

15. A voltaic cell is based on the two standard half-cell reactions:
$Ag^+(aq) + e^- \rightarrow Ag(s)$; $E° = 0.80$ V
$Mn^{2+}(aq) + 2\ e^- \rightarrow Mn(s)$; $E° = -1.18$ V
Which of the following statements is **false** ?
A) The anode half-cell reaction is $Mn(s) \rightarrow Mn^{2+}(aq) + 2\ e^-$.
B) The reducing agent is $Ag(s)$.

C) Under standard conditions, the cell potential is 1.98 V.

D) The cell potential decreases with time.

E) The oxidizing agent is $Ag^+(aq)$.

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16. Given:
$Ni^{2+}(aq) + 2\ e^- \rightarrow Ni(s)$; $E° = –0.23$ V
$2\ H^+(aq) + 2\ e^- \rightarrow H_2(g)$; $E° = 0.00$ V
$Ag^+(aq) + e^- \rightarrow Ag(s)$; $E° = 0.80$ V
$NO_3^-(aq) + 4\ H^+(aq) + 3\ e^- \rightarrow NO(g) + 2\ H_2O(l)$; $E° = 0.96$ V
Which of the following statements is true?
A) Ni^{2+} reacts spontaneously with 1 *M* $H^+(aq)$ to form H_2.
B) Ni^{2+} reacts spontaneously with $H_2(g)$.
C) Ag(*s*) reacts spontaneously with Ni^{2+}.
D) Ag(*s*) reacts spontaneously with 1 *M* $H^+(aq)$.
E) Ag(*s*) reacts spontaneously with 1 *M* NO_3^- in 1 *M* $H^+(aq)$.

17. The rate of a reaction depends on __________.
A) collision energy B) collision orientation
C) collision frequency D) all of the above

18. In which of the following aqueous solutions does the weak acid exhibit the lowest percent ionization?
A) 0.1 *M* HClO ($K_a = 3.0 \times 10^{-8}$)
B) 0.01 *M* HClO ($K_a = 3.0 \times 10^{-8}$)
C) 0.1 *M* $HC_2H_3O_2$ ($K_a = 1.8 \times 10^{-5}$)
D) 0.01 *M* $HC_2H_3O_2$ ($K_a = 1.8 \times 10^{-5}$)

19. The molar solubility of ________ is not affected by the pH of the solution.
A) MnS B) BaF_2 C) PbI_2 2 D) $CaCO_3$

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Part B: Short questions (6 points)

20. A buffer contains 0.10 mol of acetic acid and 0.13 mol of sodium acetate in 1.00 L.The K_a of acetic acid is 1.8×10^{-5}.(5 points)

(a) What is the pH of this buffer?

(b) What is the pH of the buffer after the addition of 0.02 mol of KOH?

(c) What is the pH of the buffer after the addition of 0.02 mol of HNO_3?

(a) $K_a = \dfrac{[H^+][CH_3COO^-]}{[CH_3COOH]}; [H^+] = \dfrac{K_a[CH_3COOH]}{[CH_3COO^-]}$

$$[H^+] \approx \frac{1.8 \times 10^{-5}\,(0.10)}{(0.13)} = 1.385 \times 10^{-5} = 1.4 \times 10^{-5}\text{ M; pH} = 4.86$$

(b)

$CH_3COOH(aq)$	+	$KOH(aq)$	$\rightarrow$	$CH_3COO^-(aq) + H_2O(l) + K^+(aq)$
0.10 mol		0.02 mol		0.13 mol
–0.02 mol		–0.02 mol		+0.02 mol
0.08 mol		0 mol		0.15 mol

$$[H^+] = \frac{1.8 \times 10^{-5}\,(0.08\text{ mol}/1.00\text{ L})}{(0.15\text{ mol}/1.00\text{ L})} = 9.60 \times 10^{-6} = 1 \times 10^{-5}\text{ M; pH} = 5.02 = 5.0$$

(c)

$CH_3COO^-(aq)$	+	$HNO_3(aq)$	$\rightarrow$	$CH_3COOH(aq) + NO_3^-(aq)$
0.13 mol		0.02 mol		0.10 mol
–0.02 mol		–0.02 mol		+0.02 mol
0.11 mol		0 mol		0.12 mol

$$[H^+] = \frac{1.8 \times 10^{-5}\,(0.12\text{ mol}/1.00\text{ L})}{(0.11\text{ mol}/1.00\text{ L})} = 1.96 \times 10^{-5} = 2.0 \times 10^{-5}\text{ M; pH} = 4.71$$

21. For the isothermal expansion of a gas into a vacuum, $\Delta E = 0$, $q = 0$, and $w = 0$. (**a**) Is this a spontaneous process? (**b**) Explain why no work is done by the system during this process. (**c**) What is the "driving force" for the expansion of the gas: enthalpy or entropy? (3 points)

Answer:

(**a**) Spontaneous process.

(**b**) Isothermal expansion of a gas into a vacuum P=0, w=-$P\Delta V$=0

(**c**) entropy

22. Predict the sign of ΔS_{sys} for each of the following processes: (**a**) Gaseous H_2 reacts with liquid palmitoleic acid ($C_{16}H_{30}O_2$, unsaturated fatty acid) to form liquid

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palmitic acid ($C_{16}H_{30}O_2$ saturated fatty acid). (**b**) Liquid palmitic acid solidifies at 1 °C to solid palmitic acid. (**c**) Silver chloride precipitates upon mixing $AgNO_3$(*aq*) and NaCl(*aq*). (**d**) Gaseous H_2 dissociates in an electric arc to form gaseous H atoms (used in atomic hydrogen welding) (4 points)

Answer:

(a) $\Delta S_{sys}<0$ OR negative

(b) $\Delta S_{sys}<0$ OR negative

(c) $\Delta S_{sys}<0$ OR negative

(d) $\Delta S_{sys}>0$ OR positive

23. The value of K_a for nitrous acid (HNO_2) at 25 °C is 4.5×10^{-4}. **(a)** Write the chemical equation for the equilibrium that corresponds to K_a. **(b)** By using the value of K_a, calculate $\Delta G°$ for the dissociation of nitrous acid in aqueous solution. **(c)** What is the value of ΔG at equilibrium? **(d)** What is the value of ΔG when $[H^+] = 5.0 \times 10^{-2}$ *M*, $[NO_2^-] = 6.0 \times 10^{-4}$ *M*, and $[HNO_2] = 0.20$ *M*? (4 points)

(a) $HNO_2(aq) \leftrightarrow H^+(aq) + NO_2^-(aq)$

(b) $K_a = 4.5 \times 10^{-4}$

At equilibrium $\Delta G^0 = -RTlnK_a = 8.314 \times 298 \times \ln(4.5 \times 10^{-4}) = 19.1kJ$

(c) At equilibrium $\Delta G = 0$

(d) $\Delta G = \Delta G^0 + RTlnQ$

$$\Delta G = 19.1 + 8.314 \times 298 \div 1000 \times \ln\frac{5.0 \times 10^{-2} \times 6.0 \times 10^{-4}}{0.20} = -2.8\ kJ$$

24. Cytochrome, a complicated molecule that we will represent as $CyFe^{2+}$, reacts with the air we breathe to supply energy required to synthesize adenosine triphosphate (ATP). The body uses ATP as an energy source to drive other reactions. Given:

$O_2(g) + 4\ H^+(aq) + 4\ e^- \rightarrow 2\ H_2O(l)$ $\qquad E°_{red} = +\ 0.82$ V

$CyFe^{3+}(aq) + e^- \rightarrow CyFe^{2+}(aq)$ $\qquad E°_{red} = +\ 0.22$ V

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(a) Write the balanced chemical equation (including state symbol) for the oxidation of $CyFe^{2+}$ by O_2.

(b) Calculate the standard cell potential of the overall reaction under standard condition.

(c) Calculate the value of $\Delta G°$ for the overall reaction (F = 96,485 C/ mol). (6 points)

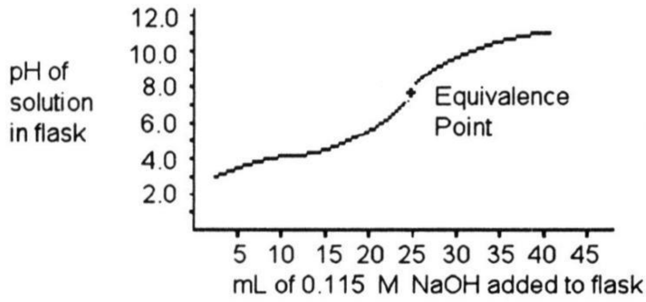
Answer:

(a) $O_2(g) + 4\ H^+(aq) + 4\ CyFe^{2+}(aq) \rightarrow 2\ H_2O(l) + 4\ CyFe^{3+}(aq)$

(b) $E^o = E^o_{red}$ (cathode) – E^o_{red} (anode) = + 0.82 V – (+ 0.22 V) = 0.60 V

(c) $\Delta G^o = -nFE^o = -4 \times 96485$ C/mol$\times$0.60 J/C = – 2.3 × 10^5 J/mol OR – 230 kJ/mol

25. A 25.0 mL sample of a solution of an unknown compound is titrated with a 0.115 M NaOH solution. The titration curve below was obtained. Estimate the pKa of this unknow compound. (2 points)



$$\text{pH} = \text{pKa} + \log_{10}\left(\frac{\text{concentration of conjugate base}}{\text{concentration of acid}}\right)$$

26. (a) Some reactions that are predicted by their sign of $\Delta G°$ to be spontaneous at room temperature do not proceed at a measurable rate at room temperature. Explain why?

(b) A suitable catalyst increases the rate of such a reaction. What effect does the catalyst have on $\Delta G°$ for the reaction? Explain. (4 points)

(a)The sign of $\Delta G°$ discusses the thermodynamic procession. It means that at a given temperature, the reaction can be or can not be spontaneous. But it doesn't means that at a given temperature, the reaction rate can be measured.

(b)The catalyst does not affect the $\Delta G°$ of the reaction. The catalyst affects the activated energy of a reaction.

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Appendices

Fundamental Physical Constants

atomic mass constant (amu)	$1.66053904 \times 10^{-27}$ kg
Avogadro's number (N_A)	$6.022140857 \times 10^{23}$ mol^{-1}
Boltzmann constant (k)	$1.3806485 \times 10^{-23}$ J·K^{-1}
electron charge (e)	$1.6021766208 \times 10^{-19}$ C
Faraday's constant (F)	96485 C·mol^{-1}
gas constant (R)	8.206×10^{-2} L·atm·mol^{-1}·K^{-1} = 8.314 J·mol^{-1}·K^{-1}
Planck's constant (h)	$6.626070040 \times 10^{-34}$ J·s
neutron mass (m_n)	$1.674927471 \times 10^{-27}$ kg = 1.008664916 amu
Electron mass (m_e)	$9.10938356 \times 10^{-31}$ kg = $5.4857990946 \times 10^{-4}$ amu
proton mass (m_p)	$1.672621898 \times 10^{-27}$ kg = 1.007276466 amu
Pi (π)	3.1425926535
Rydberg constant (R_H)	1.096776×10^{7} m^{-1}
speed of light (in vacuum) (c)	2.99792458×10^{8} m·s^{-1}

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The Periodic Table of the Elements

1 **H** Hydrogen 1.00794																	2 **He** Helium 4.003
3 **Li** Lithium 6.941	4 **Be** Beryllium 9.012182											5 **B** Boron 10.811	6 **C** Carbon 12.0107	7 **N** Nitrogen 14.00674	8 **O** Oxygen 15.9994	9 **F** Fluorine 18.9984032	10 **Ne** Neon 20.1797
11 **Na** Sodium 22.989770	12 **Mg** Magnesium 24.3050											13 **Al** Aluminum 26.981538	14 **Si** Silicon 28.0855	15 **P** Phosphorus 30.973761	16 **S** Sulfur 32.066	17 **Cl** Chlorine 35.4527	18 **Ar** Argon 39.948
19 **K** Potassium 39.0983	20 **Ca** Calcium 40.078	21 **Sc** Scandium 44.955910	22 **Ti** Titanium 47.867	23 **V** Vanadium 50.9415	24 **Cr** Chromium 51.9961	25 **Mn** Manganese 54.938049	26 **Fe** Iron 55.845	27 **Co** Cobalt 58.933200	28 **Ni** Nickel 58.6934	29 **Cu** Copper 63.546	30 **Zn** Zinc 65.39	31 **Ga** Gallium 69.723	32 **Ge** Germanium 72.61	33 **As** Arsenic 74.92160	34 **Se** Selenium 78.96	35 **Br** Bromine 79.904	36 **Kr** Krypton 83.80
37 **Rb** Rubidium 85.4678	38 **Sr** Strontium 87.62	39 **Y** Yttrium 88.90585	40 **Zr** Zirconium 91.224	41 **Nb** Niobium 92.90638	42 **Mo** Molybdenum 95.94	43 **Tc** Technetium (98)	44 **Ru** Ruthenium 101.07	45 **Rh** Rhodium 102.90550	46 **Pd** Palladium 106.42	47 **Ag** Silver 107.8682	48 **Cd** Cadmium 112.411	49 **In** Indium 114.818	50 **Sn** Tin 118.710	51 **Sb** Antimony 121.760	52 **Te** Tellurium 127.60	53 **I** Iodine 126.90447	54 **Xe** Xenon 131.29
55 **Cs** Cesium 132.90545	56 **Ba** Barium 137.327	57 **La** Lanthanum 138.9055	72 **Hf** Hafnium 178.49	73 **Ta** Tantalum 180.9479	74 **W** Tungsten 183.84	75 **Re** Rhenium 186.207	76 **Os** Osmium 190.23	77 **Ir** Iridium 192.217	78 **Pt** Platinum 195.078	79 **Au** Gold 196.96655	80 **Hg** Mercury 200.59	81 **Tl** Thallium 204.3833	82 **Pb** Lead 207.2	83 **Bi** Bismuth 208.98038	84 **Po** Polonium (209)	85 **At** Astatine (210)	86 **Rn** Radon (222)
87 **Fr** Francium (223)	88 **Ra** Radium (226)	89 **Ac** Actinium (227)	104 **Rf** Rutherfordium (261)	105 **Db** Dubnium (262)	106 **Sg** Seaborgium (263)	107 **Bh** Bohrium (262)	108 **Hs** Hassium (265)	109 **Mt** Meitnerium (266)	110 (269)	111 (272)	112 (277)	113	114				

58 **Ce** Cerium 140.116	59 **Pr** Praseodymium 140.90765	60 **Nd** Neodymium 144.24	61 **Pm** Promethium (145)	62 **Sm** Samarium 150.36	63 **Eu** Europium 151.964	64 **Gd** Gadolinium 157.25	65 **Tb** Terbium 158.92534	66 **Dy** Dysprosium 162.50	67 **Ho** Holmium 164.93032	68 **Er** Erbium 167.26	69 **Tm** Thulium 168.93421	70 **Yb** Ytterbium 173.04	71 **Lu** Lutetium 174.967
90 **Th** Thorium 232.0381	91 **Pa** Protactinium 231.03588	92 **U** Uranium 238.0289	93 **Np** Neptunium (237)	94 **Pu** Plutonium (244)	95 **Am** Americium (243)	96 **Cm** Curium (247)	97 **Bk** Berkelium (247)	98 **Cf** Californium (251)	99 **Es** Einsteinium (252)	100 **Fm** Fermium (257)	101 **Md** Mendelevium (258)	102 **No** Nobelium (259)	103 **Lr** Lawrencium (262)

1995 IUPAC masses and Approved Names from http://www.chem.qmw.ac.uk/iupac/AtWt/
masses for 107-111 from C&EN, March 13, 1995, p. 35
112 from http://www.gsi.de/z112e.html